

# Measuring Media Slant through Image Analysis

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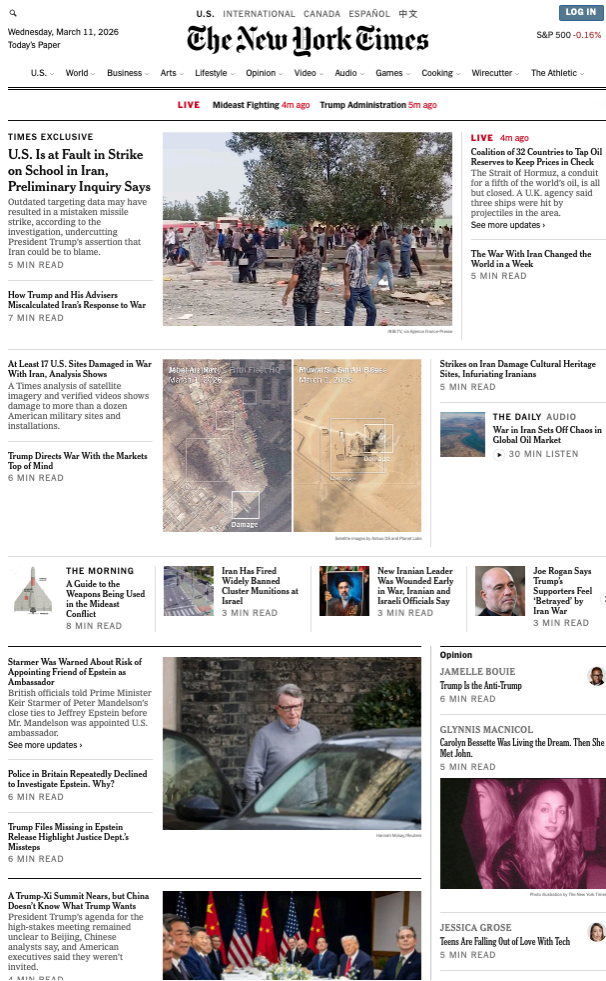
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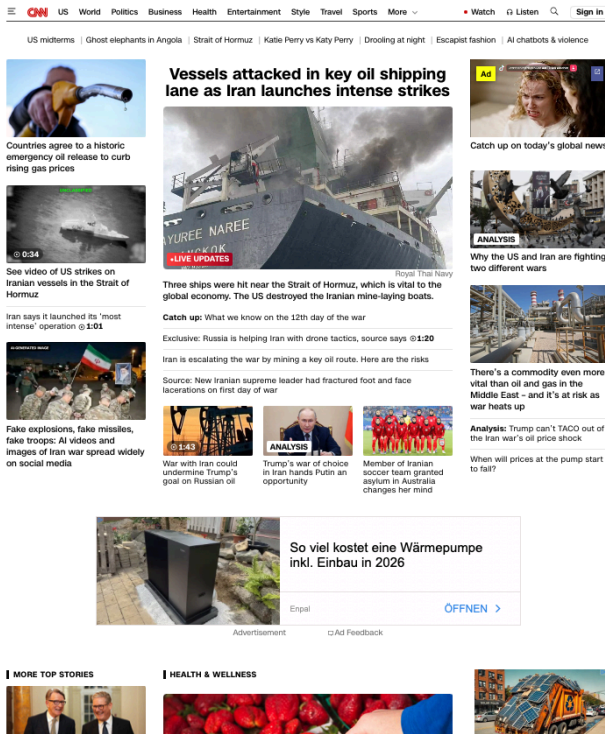
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Annual Meeting of the European Political Science Society · June 18–20, 2026 · Belfast, UK

# Images Dominate News Websites



New York Times, March 11, 2026

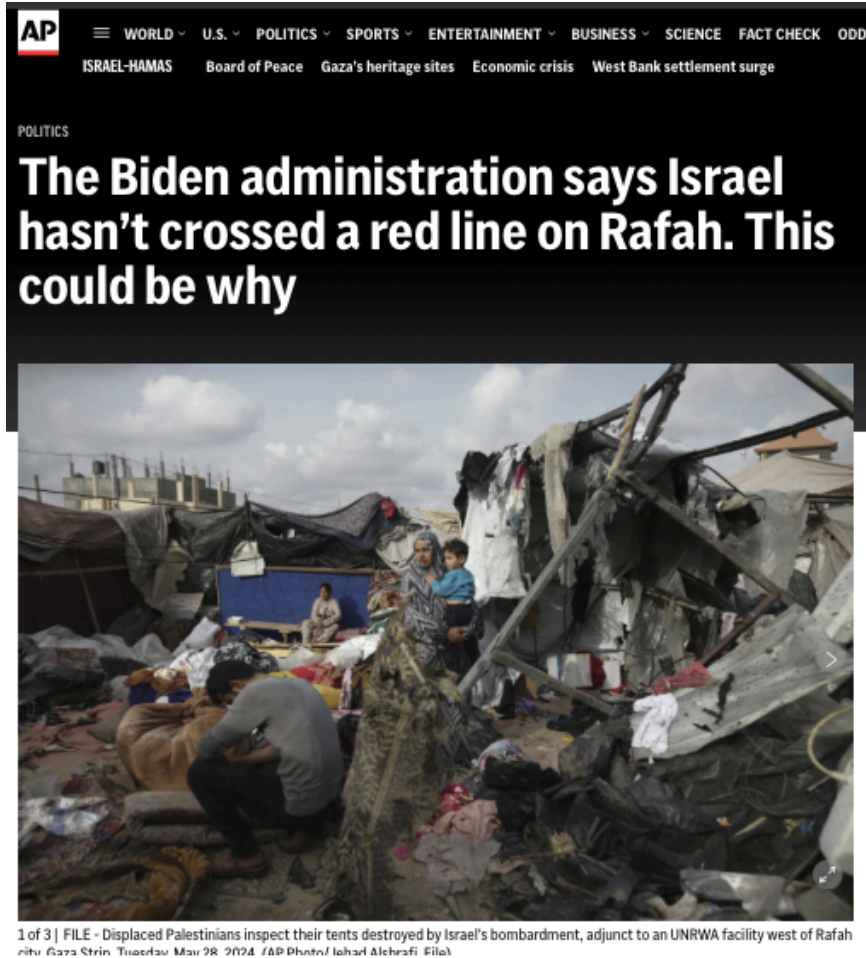


CNN, March 11, 2026



Bild, March 11, 2026

# The Same Event, Pictured Differently



AP News, June 1, 2024



Guardian, May 29, 2024

# Images Are Central in Modern Media Environments, Yet Unmeasured

- Images are omnipresent in news media, particularly online
- Images shape the perceptions of events, potentially more forcefully than text
- We know about textual media slant based, but what about visual reporting?
- Difficult to answer due to methodological hurdles

# This Study

## *Our goal:*

*Develop a method to **measure media slant** based on the **visual reporting** of media outlets on specific events.*

1. We develop a two-step, **unsupervised framework** for scaling images
2. **Application:** Visual reporting on the Gaza war, 10 outlets, Oct 2023–Oct 2024
3. **Evidence** for media slant in visual reporting, particularly for right-wing media

# Framework

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# Framework: Two Steps

***Core assumption:** News outlets that are close to each other in the latent space will, on average, use more similar news imagery.*

- 1. Measurement of image similarity:** Encode each image with a pre-trained Vision Transformer (ViT)
  - extract the [CLS] token embedding (512 dims)
  - compute pairwise **cosine similarities** across all image dyads
- 2. Statistical model to analyse image similarity:** Additive-Multiplicative Effects (AME) network model to recover latent positions of
  - (a) media outlets** based on the news imagery used by them and
  - (b) images** themselves



## Step 2: The AME Model

Additive-multiplicative effects (*Hoff 2005, 2021*) on outlet level:

$$\text{atanh}\left(\text{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^\top \vec{S}_{O_j} + \epsilon_{ij}$$

$\vec{S}_{O_i}$ : Latent position of outlet  $O$  that posted image  $i$

$\vec{S}_{O_j}$ : Latent position of outlet  $O$  that posted image  $j$

| Configuration                  | $S_i \cdot S_j$ | Predicted image similarity |
|--------------------------------|-----------------|----------------------------|
| Same pole $(+, +)$ or $(-, -)$ | $> 0$           | above baseline             |
| Opposite poles $(+, -)$        | $< 0$           | below baseline             |



# Case Study: Gaza War, 2023–2024

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# Data: Visual Reporting on the Gaza War

## Corpus

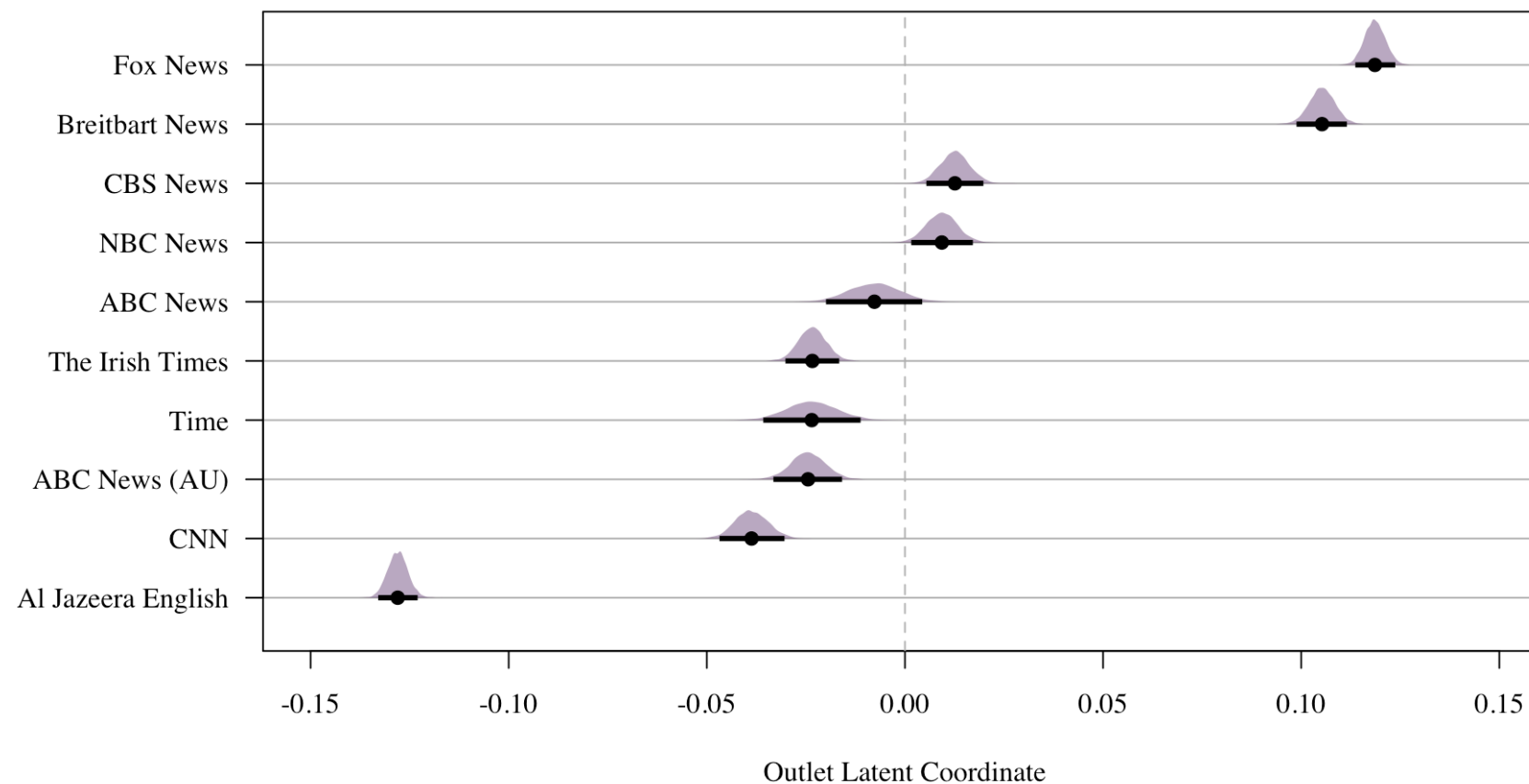
- Around 17,000 photographs from NewsAPI
- 10 news outlets: mostly US (e.g, CNN, Fox News) with some international (e.g., Al Jazeera English, Irish Times)

*Leading Question: Can we recover media slant exclusively based on news imagery?*

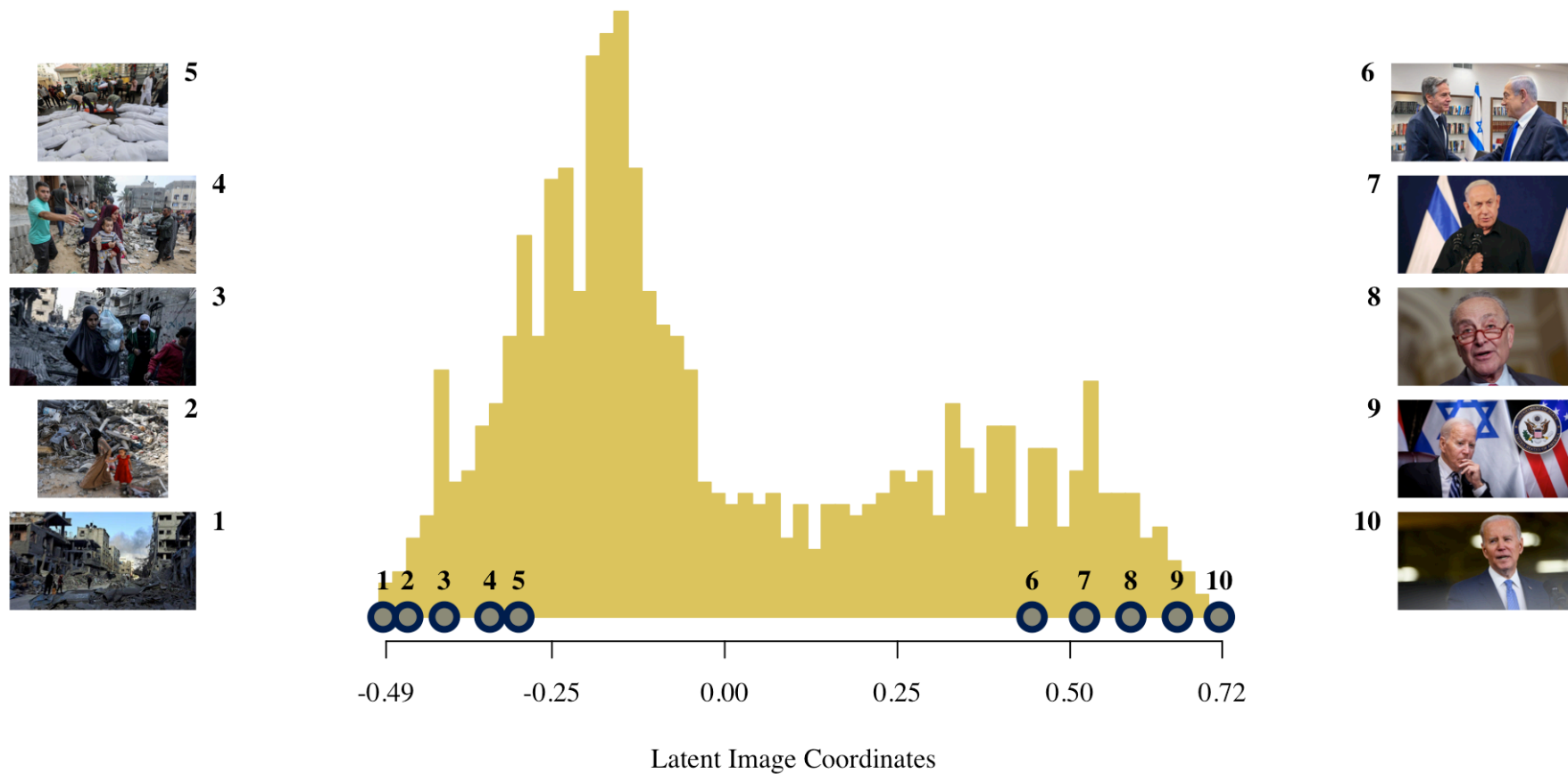
# Results

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# Outlet-Level Visual Slant



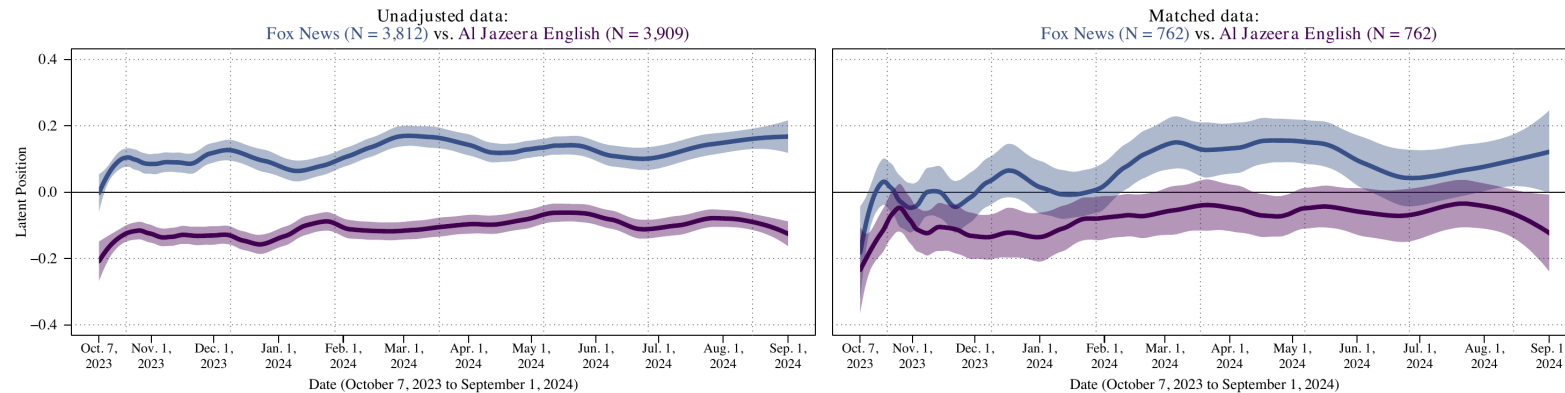
# Image-Level: What Is on the Scale?



# Is Visual Slant Just Downstream of Text Slant?

*Concern: Do outlets differ in which stories they cover (text) or in how they picture the same story?*

**Design:** within each day, match articles across two outlets on text-embedding similarity (adapting *Roberts et al. 2020*) and compare images attached to textually similar reports



# Conclusion

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# Conclusion

## Contribution

- Scalable framework for recovering latent positions from visual corpora
- Method to measure visual slant in news media

## Findings on Gaza War coverage

- Far-right outlets share a distinctive visual repertoire; Al Jazeera diverges sharply
- Scale captures a substantively meaningful dimension: elite framing vs. humanitarian suffering
- Visual slant persists after matching on text

## Measuring Media Slant through Image Analysis

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# Appendix

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## Appendix: Image-Level Model

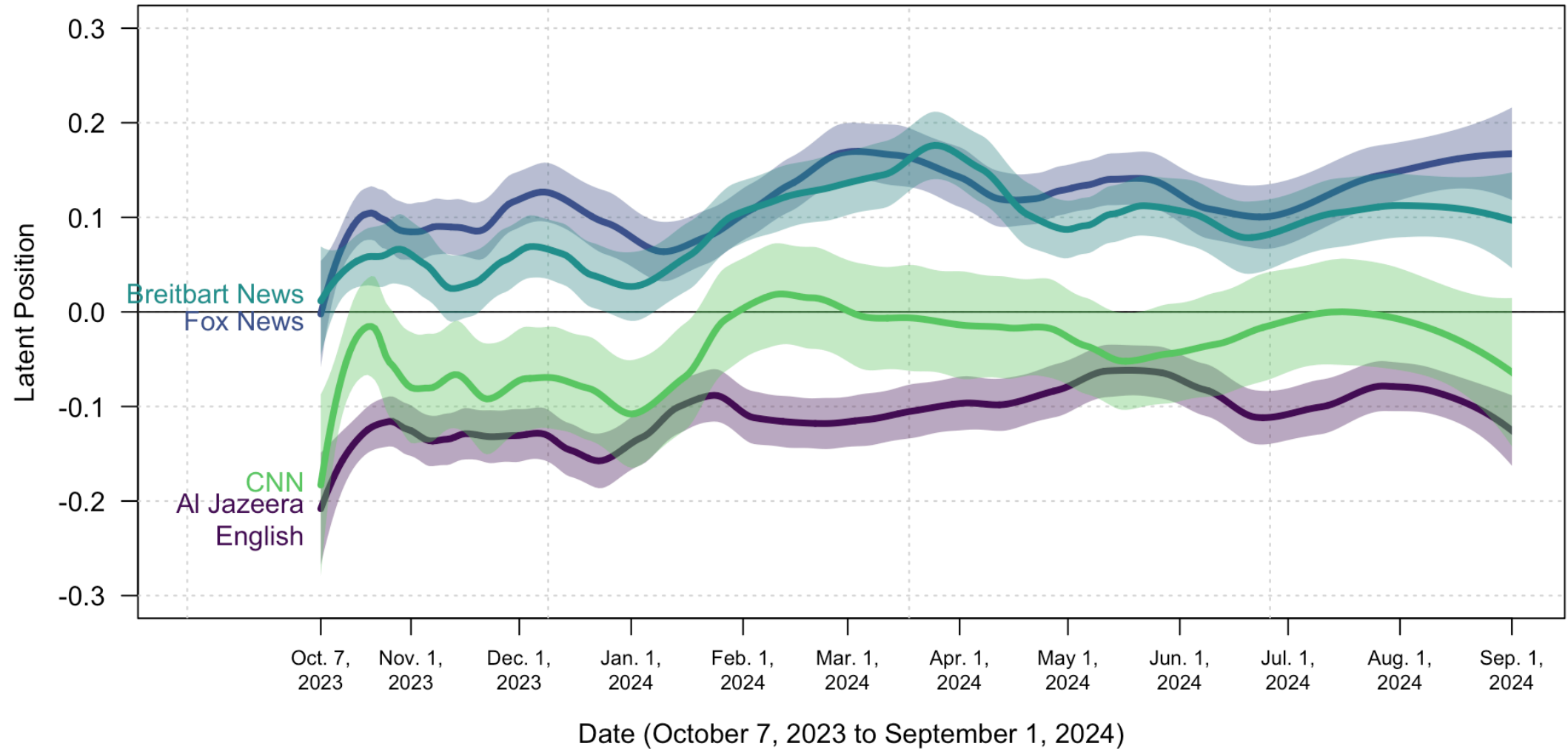
Assign a position directly to each image rather than outlets:

$$\operatorname{atanh}\left(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \vec{\mathcal{S}}_i^\top \vec{\mathcal{S}}_j + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

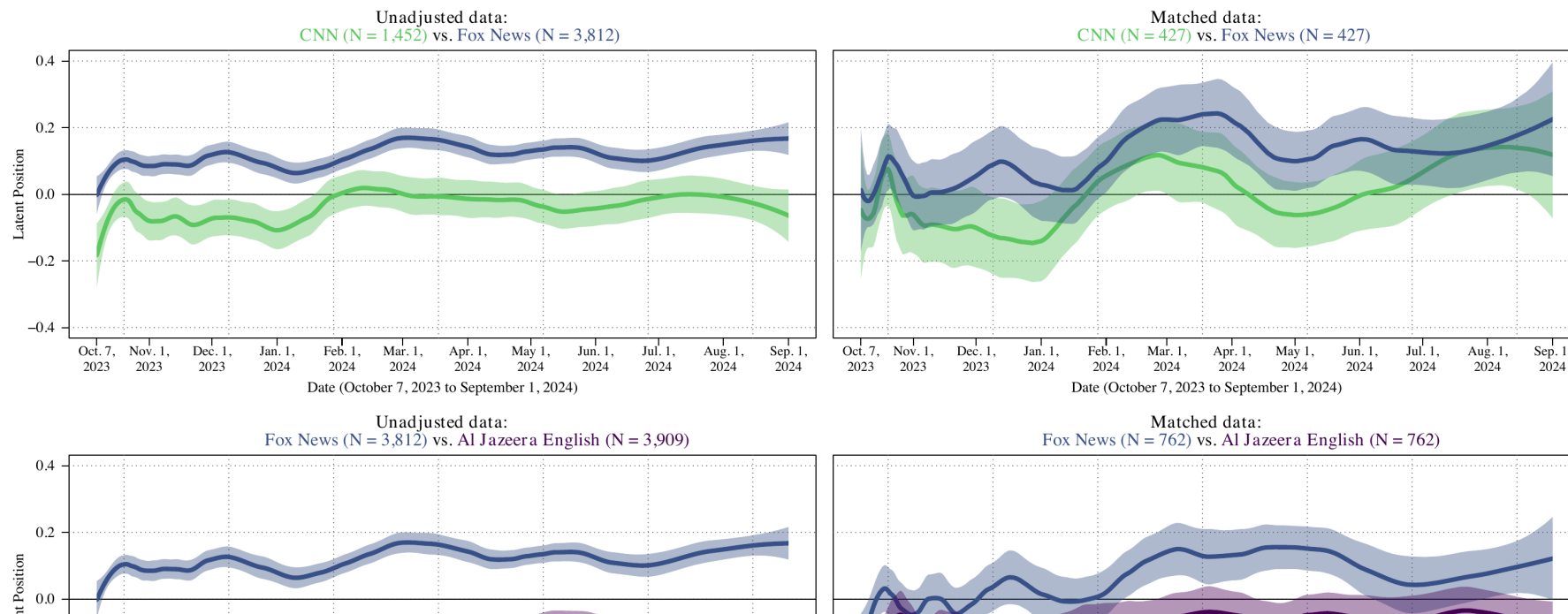
$\vec{\mathcal{S}}_i$  Latent scale position of image  $i$

- Estimated in a Bayesian framework
- No labelled data required

## Appendix: Dynamic Visual Slant: Stable Sorting, Parallel Dynamics

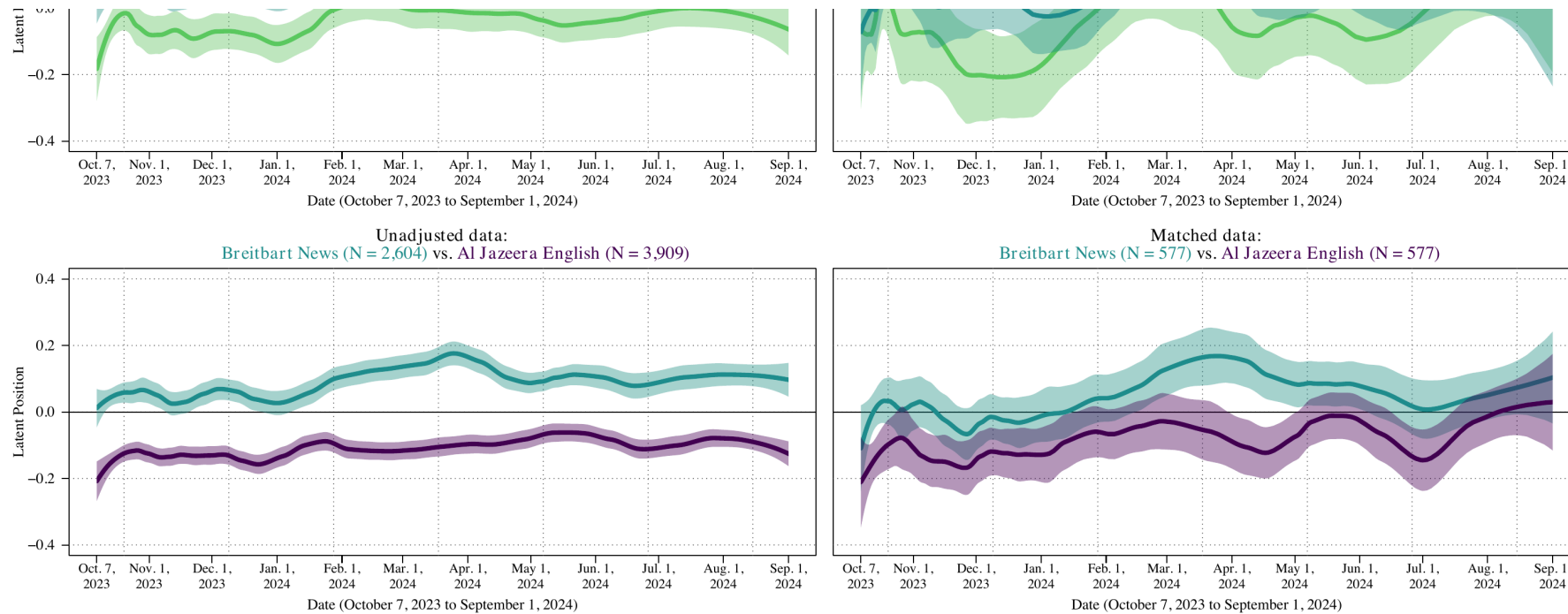


# Appendix: Visual Slant Conditional on Text (1/2)



Left: all images per outlet. Right: within-day, text-matched article pairs.

# Appendix: Visual Slant Conditional on Text (2/2)



*Left:* all images per outlet. *Right:* within-day, text-matched article pairs. Separation attenuates but persists.

# Appendix: Text-Matching Design

Adapting Roberts et al. (2020) to isolate visual framing from textual content:

1. Restrict to articles published by both outlets **on the same day**
2. Compute pairwise cosine distances on **768-dim text embeddings**
3. **Greedy 1:1 nearest-neighbour** matching across the two outlets
4. Impose a **caliper** (max permissible distance) — drop insufficiently similar pairs

*Remaining variation in image position between outlets is then attributable to **editorial image selection**, not differences in topic coverage.*

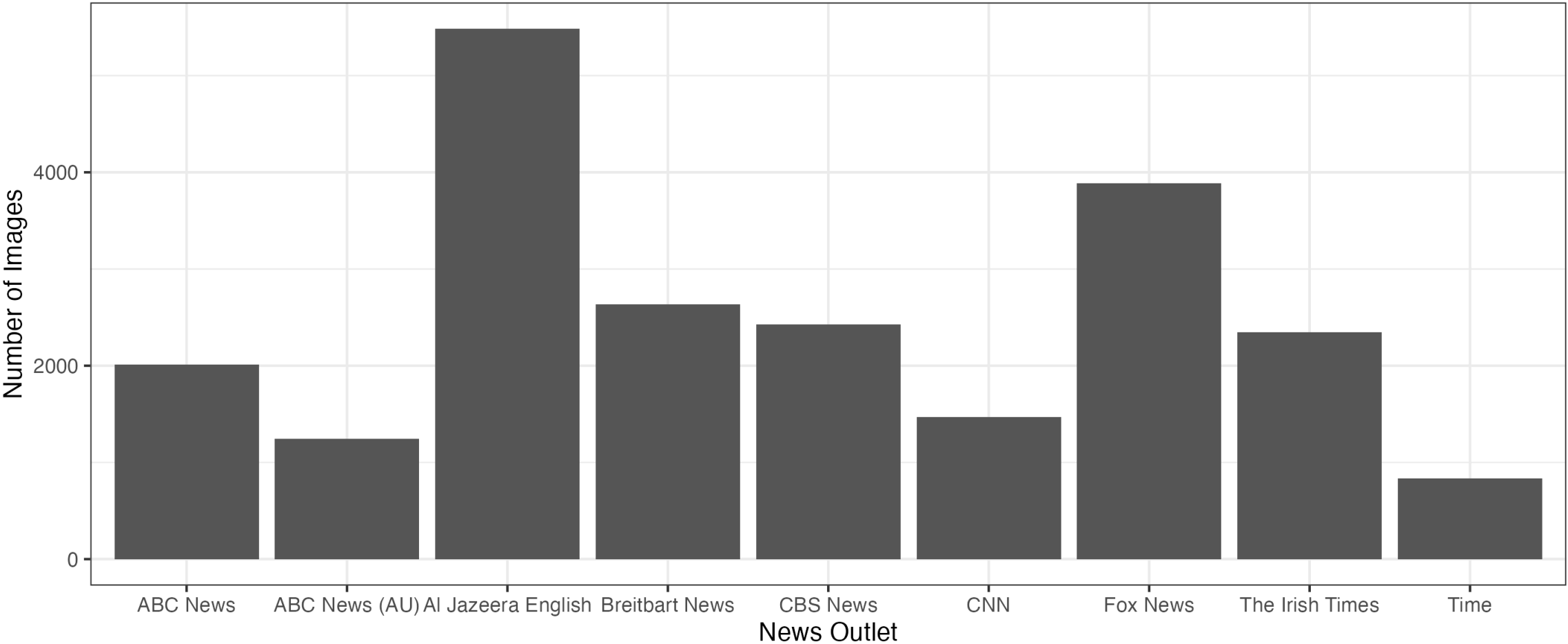


# Appendix: Text-Matched Differences (Table 1)

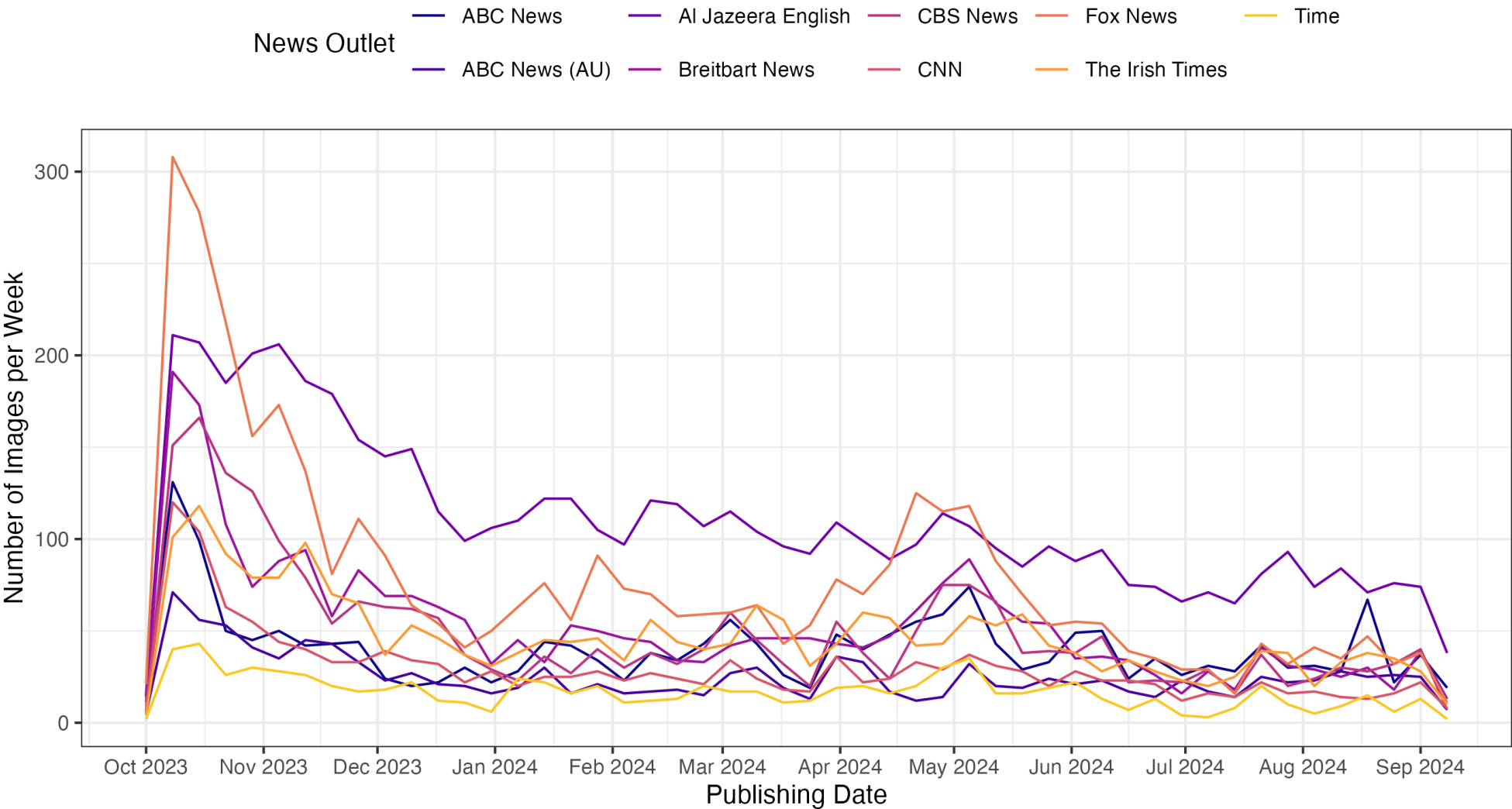
| Contrast                 | Caliper    | Est. (SE)      | $N_{\text{pairs}}$ |
|--------------------------|------------|----------------|--------------------|
| CNN vs. Fox News         | Unadjusted | 0.159 (0.009)  | —                  |
|                          | 5th pct.   | 0.102 (0.021)  | 427                |
|                          | 2.5th pct. | 0.092 (0.026)  | 274                |
|                          | 1st pct.   | 0.106 (0.032)  | 176                |
| Fox News vs. Al Jazeera  | Unadjusted | 0.224 (0.006)  | —                  |
|                          | 5th pct.   | 0.137 (0.015)  | 762                |
|                          | 1st pct.   | 0.092 (0.023)  | 287                |
| CNN vs. Breitbart        | Unadjusted | −0.128 (0.009) | —                  |
|                          | 5th pct.   | −0.112 (0.024) | 295                |
|                          | 1st pct.   | −0.122 (0.040) | 73                 |
| Breitbart vs. Al Jazeera | Unadjusted | 0.191 (0.007)  | —                  |
|                          | 5th pct.   | 0.128 (0.015)  | 577                |
|                          | 1st pct.   | 0.102 (0.025)  | 179                |

OLS of latent image position on outlet, day fixed effects, within-day matched sample.

# Appendix: Images per Outlet



# Appendix: Images per Outlet over Time



## Appendix: Step 2: The AME Model

Additive-multiplicative effects (*Hoff 2005, 2021*) — outlet level:

$$\operatorname{atanh}\left(\operatorname{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^\top \vec{S}_{O_j} + \epsilon_{ij}$$

$\mu$  is the **global baseline** similarity of news imagery across outlet pairs.

## Appendix: Step 2: The AME Model

Additive-multiplicative effects (*Hoff 2005, 2021*) — outlet level:

$$\text{atanh}\left(\text{sim}(\vec{E}_i, \vec{E}_j)\right) = \mu + \alpha_{O_i} + \alpha_{O_j} + \vec{S}_{O_i}^\top \vec{S}_{O_j} + \epsilon_{ij}$$

$\alpha_{O_i}$ : Random effects for outlet  $O$  that posted image  $i$ .

$\alpha_{O_j}$ : Random effects for outlet  $O$  that posted image  $j$ .